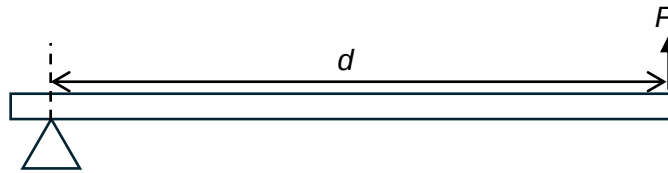


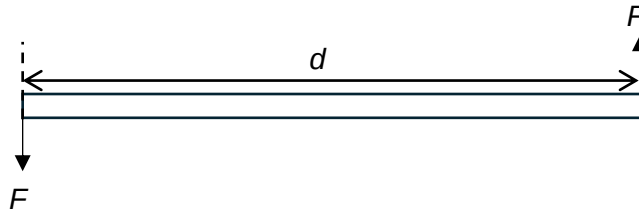
1(a) **Moment of a force:**



The moment of a force about an axis is the product of the force and the perpendicular distance [1] from the line of action of the force to the axis [1].

Minus 1 mark if no diagram given or diagram does not show corresponding d and F correctly.

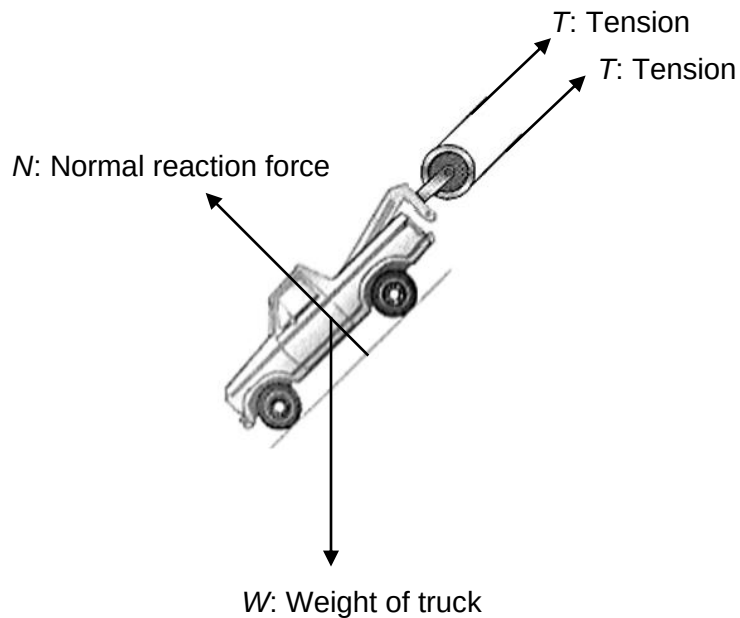
Torque of a couple:



Torque of couple is defined as the product of one force F [1] and the perpendicular distance d between the two forces [1].

Minus 1 mark if no diagram given or diagram does not show corresponding d and F correctly.

(b)(i)



(b)(ii) Considering the truck in equilibrium:

$$m_T g \sin 40^\circ = 2T \quad [1]$$

Taking moments about the centre of the pulley,

Sum of clockwise moments = sum of anti-clockwise moments

$$mg \times 3r = T \times r$$

$$mg \times 3r = \frac{m_T g \sin 40^\circ}{2} \times r \quad [1\text{m for correct clockwise expression, 1m for correct anti-clockwise expression}]$$

$$m = \frac{1500 \sin 40^\circ}{2(3)}$$

$$m = 161 \text{ kg} \quad [1]$$